BTS-V(R/S)-01-23-1377 Reg. No. [| | | | | | | |]

C

B.Tech. Degree V Semester Regular/Supplementary Examination January 2023

ME 19-205-0503 MACHINING SCIENCE AND MACHINE TOOLS

(2019 *Scheme*)

Time: 3 Hours Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: To study the geometry of single point and multi point cutting tools and mechanics of chip formation.

CO2: To get the concept of mechanics of metal cutting.

CO3: To understand the principles of working, specifications, different types and operations performed in lathe, drilling and shaping machines.

CO4: To learn the principles of working, specifications, different types and operations performed in grinding and milling machines.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Explain the formation of a built-up-edge (BUE). Explain the conditions which promote the growth of BUE.	3	L2	1	1
	(b)	How does the cutting process parameters affect the cutting tool wear in single point tools?	3	L2	2	1
	(c)	Compare 'generating' and 'forming' with reference to metal cutting machine tools.	3	L3	3	1
	(d)	What is the function served by a lead screw in a machine tool?	3	L1	3	1
	(e)	How a lathe machine is specified?	3	L1	3	1
	(f)	Describe Indian Standard wheel marking system for grinding wheels.	3	L1	4	1
	(g)	Differentiate between up milling and down milling.	3	L2	4	1
	(h)	What are the essential requirements of clamps in fixture design?	3	L2	4	1

PART B

		$(4 \times 12 = 48)$				
II.	(a)	Explain the flank wear of a cutting tool with the help of suitable sketches. Also explain its importance from the tool life point of view.	5	L2	2	1
	(b)	Find the power consumption in terms of shear and friction angles, during machining process. Use Merchant's circle diagram for derivation.	7	L4	2	2
		OR				
III.	(a)	During an orthogonal machining operation on mild steel, the results obtained are: t_1 (depth of cut) = 0.25 mm, t_2 (chip thickness) = 0.75 mm, w = 2.5 mm, $\alpha = 0^0$, Fc = 950 N and Ft = 475 N. Determine: (i) the coefficient of friction between the tool and the chip (ii) the ultimate shear stress of the work material.	6	L3	2	2
	(b)	Explain the major criteria for judging machinability.	3	L1	2	1
	(c)	Explain the relationship of tool temperature and tool life.	3	L1	2	1

(P.T.O.)

			Marks	BL	CO	PO
IV.	(a)	What are the auxiliary motions possible in a machine tool? Give examples.	5	L2	3	1
	(b)	Differentiate between shaping, slotting and planning machines with the help of block diagrams.	7	L1	3	1
		OR				
V.	(a)	Explain the application of differential mechanism in machine tools.	5	L2	3	1
	(b)	Explain the function of a translatory hydraulic drive in machine tools.	7	L2	3	1
VI.	(a)	What are the different methods used for taper turning? Explain any two.	7	L2	3	1
	(b)	A cast iron workpiece of 50 mm diameter and 200 mm length is rough turned on lathe with a tool of lead angle 20° and cutting speed of 70 m/min. Feed rate is 0.4 mm/rev. and depth of cut is 4 mm. Determine the material removal rate and machining time.	5	L3	3	2
		OR				
VII.	(a)	With a neat sketch explain the crank and slotted link quick return mechanism of shaper.	7	L1	3	1
	(b)	A plate 600 mm × 600 mm is to be machined on a shaper. Find the time required for a complete cut, if the cutting speed is 9 m/min. The return time to cutting time ratio is 1: 4 and the feed is 2 mm. The clearance at each end is 75 mm.	5	L3	3	2
VIII.	(a)	With a neat sketch explain the geometry of milling cutter.	6	L1	4	1
	(b)	Determine the time required to mill a slot in a work piece of 490 mm length with a side milling cutter of 100 mm diameter, 25 mm wide and 18 teeth. The depth of cut is 5mm and feed per tooth is 0.1 mm. Cutting speed used is 30 m/min. Assume a clearance of 2.5 mm.	6	L3	4	2
		OR				
IX.	(a)	What are the factors to be considered for the selection of grinding wheels?	6	L2	4	1
	(b)	Explain the features of an indexing jig. Give sketches.	6	L2	4	1

Bloom's Taxonomy Levels
L1 = 29.17%, L2 = 44.16%, L3 = 20.84%, L4 = 5.83%.

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